Course Name : Physics for Computer Engineers

Course Code : TE7684

Faculty : Engineering

Course Credit : 3

Course Level : 1

Sub-Committee (Specialization) : Applied Sciences

Learning Objectives :

The students will be able to:

1. Understand the central concepts and principles of quantum mechanics: the Schrodinger's equation, the wave function and its physical interpretation, and its application to quantum computing.
2. Understand and explain the application and concept of origin of band theory and transport properties of materials.
3. Understand different types of materials, their bonding, properties and applications in engineering devices.
4. Learn the principle of lasing and working of various types of lasers and understand their applications
5. Understand the phenomenon of superconductivity, properties of superconductors and their applications.

Books Recommended :

Book	Author	Publisher
Concepts of Modern Physics	A. Beiser	Mc-Graw Hill, International Edition
Fundamentals of Physics (Extended)	Halliday, Resnick and Walker,	John Wiley, 8th Ed.
Text Book of Engineering Physics	Avadhanulu & Kshirsagar	S. Chand Pub (Revised Edition).

Course Outline :

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	**Wave nature of particles and the Schrodinger equation** Introduction to Quantum mechanics, Wave nature of Particles, de-Broglie wavelength, Heisenberg Uncertainty principle, Time-dependent and time independent Schrodinger equation for wave function, Born interpretation, Particle in 1-D box, probability current, Expectation values Application of Quantum mechanics: introduction to quantum computing, Representation of Data: Qubits	**14**	**14**
2	**Electronic materials** Free electron theory, Density of states and energy band diagrams, origin of band gap, Energy bands in solids, E-k diagram, Direct and indirect bandgaps Types of electronic materials: metals, semiconductors, and insulators, Density of states, Occupation probability, Fermi level, Effective mass, Phonons.	**8**	**8**
3	**Semiconductors**	**10**	**10**

	Intrinsic and extrinsic semiconductors, Dependence of Fermi level on carrier-concentration and temperature (equilibrium carrier statistics), Carrier generation and recombination Carrier transport: drift velocity, Hall effect, Hall effect sensor, magnetic field detection, magnetic switching		
4	**Light-semiconductor interaction (LASER)** Optical transitions in bulk semiconductors: absorption, spontaneous emission, and stimulated emission; Joint density of states, Density of states for photons, Optical loss and gain Application of Lasers in security, hologram, and in medical field	**6**	**6**
5	**Superconductors** Meissner Effect, concept of cooper pairs, BCS Theory, Type I and Type II superconductors, London penetration depth Applications of superconductor: Josephson junction, superconducting qubits	**7**	**7**
	Total	**45**	**45**

Pre Requisites :

None

Evaluation :

A. Continuous Assessment

1. Unit Tests
2. Quizzes
3. Class Tests
4. Project/Presentation

B. End Semester Examination

1. Written Examination

Pedagogy :

1. Classroom teaching with practical examples
2. Assignments
3. Quizzes/ Group discussions
4. Projects/Presentations

Expert :

Dr. Meena Laad,Head and Professor,SIT

Dr. Brajesh Pandey,Associate Professor,SIT

Dr. Neeru Bhagat,Associate Professor,SIT

Dr. Rupali Nagar,Assistant Professor,SIT